THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The solution of $2x + y = 10$, $x - y = 2$ has $x =$ 1

(a) 6 **(b)** -4 **(c)** 2 **(d)** 4

2. For $x^2 - 8x + k = 0$ to have equal roots, $k =$ 1

(a) 8 **(b)** 64 **(c)** 4 **(d)** 16

3. If HCF(510, 92) = 2, then LCM(510, 92) is: 1

(a) 23460 **(b)** 46920 **(c)** 602 **(d)** 1020

4. If one zero of $x^2 - 9x + 18$ is 3, the other zero is: 1

(a) 6 **(b)** 18 **(c)** 12 **(d)** -6

5. The solution of $3x + 2y = 14$, $2x - 3y = 5$ has $y =$ 1

(a) 4 **(b)** 1 **(c)** -1 **(d)** 5

6. If one root of $x^2 - 11x + k = 0$ is 3, then k is: 1

(a) 8 **(b)** 27 **(c)** 11 **(d)** 24

7. The product of a non-zero rational and an irrational number such as $3\sqrt{5}$ is: 1

(a) sometimes rational **(b)** always an integer

(c) always irrational **(d)** always rational

8. A quadratic polynomial whose zeroes are -2 and 5 is: 1

(a) $x^2 - 3x + 10$ **(b)** $x^2 + 3x + 10$ **(c)** $x^2 - 3x - 10$ **(d)** $x^2 + 3x - 10$

9. For the pair $x + y = 14$ and $x - y = 4$, the value of xy is: 1

(a) 53 **(b)** 14 **(c)** 90 **(d)** 45

10. The roots of $x^2 - 2x - 24 = 0$ are: 1

(a) 4 and -6 **(b)** -4 and -6 **(c)** 4 and 6 **(d)** -4 and 6

11. The decimal expansion of $\frac{17}{8}$ terminates after how many places of decimals? 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

12. If α, β are zeroes of $x^2 - 5x - 24$, then $\alpha + \beta$ is: 1

(a) 5 **(b)** -5 **(c)** -24 **(d)** 6

13. The pair $x + 3y = 4$ and $3x + ky = 13$ has no solution when $k =$ 1

(a) 3 **(b)** 12 **(c)** 10 **(d)** 9

14. The discriminant of $3x^2 + 2x + 5 = 0$ is: 1

(a) -56 **(b)** -26 **(c)** 64 **(d)** 56

15. Which of the following is irrational? 1

(a) $\sqrt{4} + 2$ **(b)** $\sqrt{49} - 2$ **(c)** $\sqrt{7} + 2$ **(d)** $2 \times \sqrt{4}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the value of k for which the pair $x + 2y = 3$, $2x + ky = 7$ has no solution. 2

17. Find k for which $x^2 - 8x + k = 0$ has two equal roots. 2

18. Check whether 14^n can end with the digit 0 for any natural number n. 2

19. The graph of $y = p(x)$ cuts the x-axis at $x = 1$ and $x = 4$ only. What are the zeroes of $p(x)$, and what is its likely degree? 2

20. A man tells his daughter, "Seven years ago I was 6 times as old as you were. Three years from now I shall be 3 times as old as you will be." Form the pair of linear equations (do not solve). 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. On comparing the ratios, find whether the lines $9x + 3y = 12$ and $18x + 6y = 24$ intersect, are parallel or coincide. 3

22. Find the roots of $x^2 - 6x + 4 = 0$ using the quadratic formula (if they exist). 3

23. Show that $2 + 3\sqrt{7}$ is irrational, given that $\sqrt{7}$ is irrational. 3

24. If 3 is a zero of the polynomial $x^2 + kx + 12$, find k and the other zero. 3

OR

Solve by elimination method: $3x + 2y = 14$ and $2x - 3y = 5$. 3

25. Find the roots of $x^2 - 2x - 15 = 0$ using the quadratic formula (if they exist). 3

OR

Find the HCF and LCM of 510 and 92 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Prove that $\sqrt{19}$ is irrational. 5

OR

If α, β are the zeroes of $x^2 - 3x - 4$, form a quadratic polynomial whose zeroes are $\alpha + 1$ and $\beta + 1$. 5

27. A father is 7 times as old as his son. After 6 years the father will be 4 times as old as the son will be. Find their present ages. 5

OR

A train travels 420 km at a uniform speed. If the speed had been 10 km/h more, it would have taken 1 hour less for the journey. Find the original speed. 5

END OF QUESTION PAPER **Total: 50 Marks**

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